The Divergence–Period Symmetry Realized on the Zeta Series: One Conjugation, Two Eigenspaces (Exercise XXXVI)

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Abstract

We formalize, with five machine-checked theorems, the way in which a single conjugation symmetry splits the terms of the zeta series into two eigenspaces. Each term $1/n^s$ factors as $n^{-\sigma} \cdot e^{-it\ln n}$, where $s = \sigma + it$: the divergence axis (the real part) controls the modulus, $\|n^s\| = n^{\sigma}$, and the period axis (the imaginary part) contributes a unit modulus that never affects convergence. The two axes are the two eigenspaces of the same symmetry; each axis has its own mirror pairs that collapse to the unit 1: $e^{i\theta} \cdot e^{-i\theta} = 1$ on the period axis, and $r \cdot (1/r) = 1$ on the divergence axis. Conjugation is the reversal of the period axis, $\overline{n^s} = n^{\overline{s}}$ for positive real n. The functional equation $\zeta(1-s) = 2(2\pi)^{-s}\Gamma(s)\cos(\pi s/2)\zeta(s)$ pairs the divergent region with the convergent region through this same symmetry — it is the mechanism of analytic continuation, and it is already a machine-checked theorem of mathlib (`riemannZeta_one_sub`). Zero errors, zero `sorry`.

Keywords: Lean, formalization, zeta function, functional equation, analytic continuation, conjugation symmetry, 0pat

1 The series term as a product of two eigenspaces

The zeta function is born as a series, and each term of the series is a product of two factors. For a positive integer n and a complex exponent $s = \sigma + it$,

$$n^s = n^{\sigma+it} = n^{\sigma} \cdot n^{it} = n^{\sigma} \cdot e^{it\ln n}.$$

The first factor n^{σ} is real; it grows or decays with the real part of s. The second factor $e^{it\ln n}$ lies on the unit circle; it oscillates with the imaginary part of s. The series term $1/n^s = n^{-\sigma} \cdot e^{-it\ln n}$ is therefore the product of a *divergence factor* (real, monotone in σ) and a *period factor* (unit modulus, oscillating in t).

The first theorem makes this split precise and machine-checked.

The modulus of a term is the divergence factor (`zeta_term_norm_split`). For $n > 0$ and $s \in \mathbb{C}$,

$$\|n^s\| = n^{\operatorname{Re} s}.$$

The modulus of the term is entirely controlled by the divergence axis (the real part of s). The period axis never affects the modulus: $|e^{-it\ln n}| = 1$ for every t. In this sense the two axes play asymmetric roles in convergence — but they are not independent structures; the next section shows that both are eigenspaces of one and the same conjugation symmetry.

2 One symmetry, two mirror pairs

Conjugation $z \mapsto \overline{z}$ fixes the real axis pointwise and reverses the imaginary axis. The two axes of the series term are the two eigenspaces of this single symmetry: a real number is fixed (eigenvalue $+1$), a pure imaginary number is negated (eigenvalue -1). Each eigenspace carries its own mirror pairs, and on both axes a mirror pair collapses to the unit 1.

The period-axis mirror pair (`period_pair_reduces`). The pair $e^{i\theta}$ and $e^{-i\theta}$ are conjugate: for real θ, $\overline{e^{i\theta}} = e^{-i\theta}$. Their product is the unit,

$$e^{i\theta} \cdot e^{-i\theta} = e^0 = 1.$$

The exponent coordinate is negated, and the two mirror terms annihilate to the neutral element of multiplication. This is the algebraic identity $\exp(\theta) \cdot \exp(-\theta) = 1$, machine-checked for complex θ.

The divergence-axis mirror pair (`divergence_pair_reduces`). The pair r and $1/r$ is not a conjugate pair in the analytic sense — both numbers are real — but it is the mirror pair of the divergence axis: under the logarithm the two coordinates are negatives of each other, $\ln(1/r) = -\ln r$. Their product is the unit,

$$r \cdot \frac{1}{r} = 1 \qquad (r \neq 0).$$

The logarithmic coordinate is negated, and the two mirror terms annihilate to the neutral element of multiplication, exactly as on the period axis.

The two pairs are the two faces of one symmetry. On the period axis the mirror relation is $\theta \mapsto -\theta$ (via conjugation); on the divergence axis it is $\ln r \mapsto -\ln r$ (via reciprocation). In both cases a single coordinate is negated and the product of the mirror pair is 1. The two theorems `period_pair_reduces` and `divergence_pair_reduces` verify both faces: the same collapse, on the two eigenspaces of the same symmetry.

3 Conjugation is the reversal of the period axis

If the series term n^s is a product of a divergence factor and a period factor, then conjugation acts on it through the period factor alone. Conjugating the whole term is the same as conjugating the exponent:

Conjugation passes through the power (`term_conj_symmetry`). For $n > 0$ and $s \in \mathbb{C}$,

$$\overline{n^s} = n^{\overline{s}}.$$

Since n is positive real, its argument is 0, not π; mathlib's `Complex.conj_cpow` then moves the conjugation across the power. The statement is exactly the eigenspace behavior of the symmetry: $\overline{s} = \sigma - it$ fixes the real part (the divergence axis, σ unchanged) and negates the imaginary part (the period axis, $t \mapsto -t$). Conjugation reverses the period axis and leaves the divergence axis untouched.

4 The same symmetry pairs the two regions

The split of the term into two eigenspaces is not only a local observation about a single term. The same symmetry connects the region where the series diverges with the region where it converges.

The functional equation is the pairing (`functional_equation_bridges`). For $s \neq 1$ and $s \neq -n$ for all $n \in \mathbb{N}$,

$$\zeta(1-s) = 2\,(2\pi)^{-s}\,\Gamma(s)\,\cos\!\left(\frac{\pi s}{2}\right)\,\zeta(s).$$

This is the functional equation of the zeta function, in the exact form of mathlib's `riemannZeta_one_sub`; the theorem above is that statement, machine-checked. The map $s \mapsto 1-s$ sends the convergent region ($\operatorname{Re} s > 1$) to the divergent region ($\operatorname{Re} s < 0$): the values of the divergent series are obtained from the convergent region through this symmetry. The pairing $s \leftrightarrow 1-s$ is the same conjugation symmetry of the series terms acting on the exponent: $\sigma \mapsto 1-\sigma$ negates the divergence coordinate around its fixed point, exactly as $t \mapsto -t$ negates the period coordinate. The mechanism by which the divergent series acquires its values is the mechanism by which the two axes were mirrored in the first place.

5 Honest boundary and position

The content is classical: the factorization of n^s into a real power and a unit-modulus phase, the two mirror identities $e^{i\theta}e^{-i\theta} = 1$ and $r(1/r) = 1$, conjugation across the power for positive real bases, and the zeta functional equation. All five theorems are known mathematics, and the fifth is quoted directly from mathlib rather than proved anew. The value of the exercise is the precise statement of the structure: the divergence axis and the period axis of the series term are the two eigenspaces of one conjugation symmetry, and the functional equation is that same symmetry pairing the divergent region with the convergent region — analytic continuation as the symmetry's mechanism. No claim is made beyond the five machine-checked statements; the reader is invited to inspect the artifact.

Artifact

The artifact contains `proof.lean` (Lean 4.32.2, mathlib v4.32.2): five theorems, zero errors, zero `sorry`. Build: `lean proof.lean`.